

# Function Transformations — Solutions

Complete Step-by-Step Solutions with Trap Alerts & Key Insights

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## Question 1 [Easy]

**Correct Answer: A) (2, 8)**

**Step-by-Step Solution:**

- $g(x) = f(x) + 5$  shifts the graph up by 5.
- Point (2, 3) on  $f$  becomes  $(2, 3+5) = (2, 8)$  on  $g$ .

**Why each wrong option fails:**

- B) (7, 3) — Shifted  $x$  by 5 instead of  $y$ .
- C) (2, -2) — Subtracted 5 instead of adding.
- D) (7, 8) — Shifted both  $x$  and  $y$  by 5.

■ ■ **TRAP ALERT:** Adding to  $f(x)$  shifts vertically ( $y$ -values), NOT horizontally.

■ **KEY INSIGHT:**  $f(x) + k$  shifts up by  $k$ .  $f(x + k)$  shifts LEFT by  $k$ .

■ **FAST METHOD:**  $y$ -value goes from 3 to  $3 + 5 = 8$ .  $x$  stays the same.

■ 🏠 **VERDICT:** Hand-solve — direct application of vertical shift rule.

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## Question 2 [Easy]

**Correct Answer: B)  $h(x) = f(x - 4)$**

**Step-by-Step Solution:**

- Translating right by 4 means replacing  $x$  with  $(x - 4)$ .
- $h(x) = f(x - 4)$ .

**Why each wrong option fails:**

- A)  $f(x + 4)$  — This shifts LEFT by 4.
- C)  $f(x) + 4$  — This shifts UP by 4.
- D)  $f(x) - 4$  — This shifts DOWN by 4.

■ ■ **TRAP ALERT:** Horizontal shifts are counterintuitive: right = subtract, left = add.

■ **KEY INSIGHT:** Right  $h$  units  $\rightarrow f(x - h)$ . Left  $h$  units  $\rightarrow f(x + h)$ . It's 'opposite.'

■ **FAST METHOD:** Right = minus inside. Left = plus inside.

■ 🏠 **VERDICT:** Hand-solve — rule recall.

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### Question 3 [Easy]

**Correct Answer: B)  $(x + 3)^2 - 2$**

#### Step-by-Step Solution:

- Left 3: replace  $x$  with  $(x + 3) \rightarrow (x + 3)^2$
- Down 2: subtract 2  $\rightarrow (x + 3)^2 - 2$

#### Why each wrong option fails:

- A)  $(x - 3)^2 - 2$  — Shifted right instead of left.
- C)  $(x - 3)^2 + 2$  — Both directions wrong.
- D)  $(x + 3)^2 + 2$  — Shifted up instead of down.

■ **TRAP ALERT:** Left means +3 inside the function (counterintuitive sign).

■ **KEY INSIGHT:** Horizontal: opposite sign. Vertical: same sign.

■ **FAST METHOD:** Left 3 =  $(x+3)$ . Down 2 = subtract 2.

■ **VERDICT:** Hand-solve — apply rules directly.

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### Question 4 [Easy]

**Correct Answer: A) Reflected over the x-axis**

#### Step-by-Step Solution:

- $-f(x)$  negates all  $y$ -values, which reflects the graph over the  $x$ -axis.

#### Why each wrong option fails:

- B)  $f(-x)$  reflects over the  $y$ -axis, not  $-f(x)$ .
- C) A shift down would be  $f(x) - 1$ , not  $-f(x)$ .
- D)  $-f(-x)$  rotates  $180^\circ$  about the origin, but we only have  $-f(x)$ .

■ **TRAP ALERT:** Confusing  $-f(x)$  ( $x$ -axis reflection) with  $f(-x)$  ( $y$ -axis reflection).

■ **KEY INSIGHT:**  $-f(x)$  flips  $y$ -values  $\rightarrow x$ -axis reflection.  $f(-x)$  flips  $x$ -values  $\rightarrow y$ -axis reflection.

■ **FAST METHOD:** Negative outside =  $x$ -axis reflection. Negative inside =  $y$ -axis reflection.

■ **VERDICT:** Hand-solve — conceptual.

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### Question 5 [Easy]

Correct Answer: A) (3, -2)

#### Step-by-Step Solution:

- Shifting up 5 adds 5 to the y-coordinate of the vertex.
- New vertex:  $(3, -7 + 5) = (3, -2)$ .

#### Why each wrong option fails:

- B) (8, -7) — Added 5 to x instead of y.
- C) (3, -12) — Subtracted 5 instead of adding.
- D) (-2, -7) — Subtracted 5 from x.

■ TRAP ALERT: 'Up 5' only affects the y-coordinate of the vertex.

■ KEY INSIGHT: Vertical shifts change k in  $(h, k)$ . Horizontal shifts change h.

■ FAST METHOD: Vertex y:  $-7 + 5 = -2$ . New vertex: (3, -2).

■ VERDICT: Hand-solve — one addition.

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### Question 6 [Medium]

Correct Answer: B) 10

#### Step-by-Step Solution:

- $q(3) = f(2 \cdot 3) = f(6) = 10$ .

#### Why each wrong option fails:

- A) 5 — Divided  $f(6)$  by 2 instead of just evaluating.
- C) 20 — Multiplied  $f(6)$  by 2.
- D) 6 — Evaluated  $f(3)$  instead of  $f(2 \cdot 3)$ .

■ TRAP ALERT:  $f(2x)$  compresses the input, not the output. The y-value doesn't change if you evaluate correctly.

■ KEY INSIGHT:  $q(3) = f(2 \cdot 3) = f(6)$ . The 2 only affects what goes INTO f.

■ FAST METHOD: Plug  $x = 3$ :  $q(3) = f(6) = 10$ .

■ VERDICT: Hand-solve — substitution.

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### Question 7 [Medium]

Correct Answer: D) 2

#### Step-by-Step Solution:

- $f(x) - 7$  shifts the entire graph down 7.
- Maximum of  $f(x)$  is 9, so maximum of  $f(x) - 7$  is  $9 - 7 = 2$ .

#### Why each wrong option fails:

- A) 16 — Added 7 instead of subtracting.
- B) 9 — Ignored the transformation.
- C) 7 — Used the shift value as the answer.

■ TRAP ALERT: The maximum value shifts by -7, not the x-coordinate.

■ KEY INSIGHT: Vertical shifts directly add to or subtract from max/min values.

■ FAST METHOD: Max of  $f - 7 = 9 - 7 = 2$ .

■ VERDICT: Hand-solve — one subtraction.

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### Question 8 [Medium]

Correct Answer: A) (8, 2)

#### Step-by-Step Solution:

- $g(x) = f(x - 3) + 2$ : shift right 3, shift up 2.
- x-intercept (5, 0) of  $f$  maps to:
- x:  $5 + 3 = 8$ , y:  $0 + 2 = 2$ .
- Point becomes (8, 2).

#### Why each wrong option fails:

- B) (2, 2) — Shifted left instead of right.
- C) (5, 2) — Only applied the vertical shift.
- D) (8, 0) — Only applied the horizontal shift.

■ TRAP ALERT: Must apply BOTH shifts: right 3 AND up 2.

■ KEY INSIGHT:  $f(x - h) + k$ : every point (a, b) on  $f$  maps to (a + h, b + k) on  $g$ .

■ FAST METHOD:  $(5+3, 0+2) = (8, 2)$ .

■ VERDICT: Hand-solve — point transformation rule.

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### Question 9 [Medium]

Correct Answer: A) Right 4, up 1

Step-by-Step Solution:

- $|x - 4| + 1$ : the  $-4$  inside shifts right 4, the  $+1$  outside shifts up 1.

Why each wrong option fails:

- B) Left 4 — Horizontal shifts are opposite to the sign inside.
- C) Down 1 —  $+1$  shifts up, not down.
- D) Both directions wrong.

■ ■ TRAP ALERT: The minus sign inside the absolute value means RIGHT, not left.

■ KEY INSIGHT:  $|x - h| + k$ : vertex moves from  $(0,0)$  to  $(h, k)$ .

■ FAST METHOD: Vertex of  $|x - 4| + 1$  is at  $(4, 1) \rightarrow$  right 4, up 1.

■ 🏠 VERDICT: Hand-solve — read directly from the expression.

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### Question 10 [Medium]

Correct Answer: A)  $(-5, 4)$

Step-by-Step Solution:

- $f(x) = (x - 1)^2 + 4$  has vertex at  $(1, 4)$ .
- $g(x) = f(x + 6) = ((x + 6) - 1)^2 + 4 = (x + 5)^2 + 4$ .
- Vertex of  $g$ :  $(-5, 4)$ .

Why each wrong option fails:

- B)  $(7, 4)$  — Shifted right instead of left.
- C)  $(-5, 10)$  — Added 6 to the y-coordinate too.
- D)  $(7, 10)$  — Both wrong.

■ ■ TRAP ALERT:  $f(x + 6)$  shifts LEFT 6, not right. The vertex  $x$  moves from 1 to  $1 - 6 = -5$ .

■ KEY INSIGHT:  $f(x + c)$  shifts the graph LEFT by  $c$  units.

■ FAST METHOD: Vertex  $x$ -coord:  $1 - 6 = -5$ .  $y$ -coord stays 4.

■ 🏠 VERDICT: Hand-solve — vertex shift.

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### Question 11 [Medium]

Correct Answer: B) [-6, 24]

#### Step-by-Step Solution:

- $h(x) = 3f(x)$  multiplies all y-values by 3.
- Range of  $f$ :  $[-2, 8]$ .
- Range of  $h$ :  $[3(-2), 3(8)] = [-6, 24]$ .

#### Why each wrong option fails:

- A)  $[-2, 8]$  — Ignored the vertical stretch.
- C)  $[-6, 8]$  — Only multiplied the minimum.
- D)  $[1, 11]$  — Added 3 instead of multiplying.

■ ■ TRAP ALERT: Vertical stretch multiplies BOTH endpoints of the range.

■ KEY INSIGHT:  $af(x)$  multiplies the range by  $|a|$ , with sign consideration.

■ FAST METHOD: Multiply range bounds by 3:  $[-6, 24]$ .

■ 🏠 VERDICT: Hand-solve — multiply both endpoints.

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### Question 12 [Medium]

Correct Answer: A) (3, 5)

#### Step-by-Step Solution:

- $g(x) = f(-x)$  reflects over the y-axis.
- Point  $(-3, 5)$  on  $f$  maps to  $(3, 5)$  on  $g$ .
- The x-coordinate negates; the y-coordinate stays.

#### Why each wrong option fails:

- B)  $(-3, -5)$  — Reflected over x-axis instead.
- C)  $(3, -5)$  — Reflected over both axes.
- D)  $(-3, 5)$  — No change (incorrect).

■ ■ TRAP ALERT:  $f(-x)$  changes the x-sign, not the y-sign.

■ KEY INSIGHT:  $f(-x)$ : y-axis reflection.  $-f(x)$ : x-axis reflection.  $-f(-x)$ : 180° rotation.

■ FAST METHOD: y-axis reflection: negate x only.  $(-3, 5) \rightarrow (3, 5)$ .

■ 🏠 VERDICT: Hand-solve — conceptual.

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### Question 13 [Medium]

Correct Answer: 6

**Step-by-Step Solution:**

- $f(x - 2)$  when  $x = 3$ :  $f(3 - 2) = f(1)$ .
- From the table,  $f(1) = 6$ .

■ ■ **TRAP ALERT:** Common error: looking up  $f(3) = -2$  instead of  $f(3-2) = f(1) = 6$ .

■ **KEY INSIGHT:** Substitute the  $x$ -value FIRST, then look up the result in the table.

■ **FAST METHOD:**  $f(3 - 2) = f(1) = 6$ . Two steps.

■ 🏠 **VERDICT:** Hand-solve — substitution + table lookup.

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### Question 14 [Medium]

Correct Answer: B) 3

**Step-by-Step Solution:**

- $g(x) = 2(x - 6)^2 + 3$ .
- The squared term is always  $\geq 0$ , so minimum of  $2(x-6)^2$  is 0.
- Minimum of  $g(x) = 0 + 3 = 3$ .
- The vertical stretch factor (2) doesn't change the minimum value because the minimum occurs when the squared term is 0.

**Why each wrong option fails:**

- A) 1.5 — Divided 3 by 2.
- C) 6 — Multiplied 3 by 2.
- D) 9 — Multiplied 3 by 3.

■ ■ **TRAP ALERT:** Students think multiplying by 2 changes the minimum, but the minimum occurs when the squared term is 0, so the  $k$ -value is unchanged.

■ **KEY INSIGHT:** In  $a(x-h)^2 + k$ , the minimum (or maximum) value is always  $k$ , regardless of  $a$ .

■ **FAST METHOD:** Minimum of  $a(x - h)^2 + k$  is  $k$ . Answer: 3.

■ 🏠 **VERDICT:** Hand-solve — vertex form reading.

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### Question 15 [Hard]

Correct Answer: A) [-2, 6]

#### Step-by-Step Solution:

- $g(x) = -f(x - 2) + 1$ .
- The domain of  $f$  is  $[-4, 4]$ , so  $f(x - 2)$  requires  $-4 \leq x - 2 \leq 4$ .
- $-4 + 2 \leq x \leq 4 + 2 \rightarrow -2 \leq x \leq 6$ .
- The reflection  $(-)$  and vertical shift  $(+1)$  don't affect the domain.
- Domain of  $g$ :  $[-2, 6]$ .

#### Why each wrong option fails:

- B)  $[-6, 2]$  — Shifted left instead of right.
- C)  $[-4, 4]$  — Ignored the horizontal shift.
- D)  $[-3, 5]$  — Used wrong shift amount.

■ ■ **TRAP ALERT:** Only horizontal transformations (inside the function) affect the domain.

■ **KEY INSIGHT:**  $f(x - h)$  shifts domain right by  $h$ . Vertical changes don't affect domain.

■ **FAST METHOD:** Domain shifts right 2:  $[-4+2, 4+2] = [-2, 6]$ .

■ 🏠 **VERDICT:** Hand-solve — domain shift rule.

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### Question 16 [Hard]

Correct Answer: 5

#### Step-by-Step Solution:

- $f(x) = 2x^2 - 8x + 11 = 2(x^2 - 4x) + 11 = 2(x - 2)^2 - 8 + 11 = 2(x - 2)^2 + 3$ .
- $f$  has minimum at  $x = 2$ .
- $g(x) = f(x - 3)$  shifts right by 3.
- $g$  has minimum at  $x = 2 + 3 = 5$ .

■ ■ **TRAP ALERT:** Must complete the square to find the vertex of  $f$  before applying the shift.

■ **KEY INSIGHT:**  $f(x - 3)$  shifts the minimum  $x$ -value right by 3:  $2 + 3 = 5$ .

■ **FAST METHOD:** Vertex of  $f$  at  $x = -b/(2a) = 8/4 = 2$ . Shift right 3:  $x = 5$ .

■ 🏠 **VERDICT:** Hand-solve — vertex formula + shift.

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### Question 17 [Hard]

Correct Answer: A) (0, 3)

#### Step-by-Step Solution:

- y-intercept means  $x = 0$ .
- $g(0) = f(3 \cdot 0) + 7 = f(0) + 7 = -4 + 7 = 3$ .
- y-intercept of  $g$ : (0, 3).

#### Why each wrong option fails:

- B) (0, -12) — Multiplied  $f(0)$  by 3 instead of using  $f(3 \cdot 0)$ .
- C) (0, 17) — Used  $|f(0)| + 7$  or some other error.
- D) (0, -4) — Ignored the +7 shift.

■ ■ TRAP ALERT:  $f(3x)$  at  $x = 0$  is  $f(0)$ , not  $3f(0)$ . The horizontal compression doesn't change  $f(0)$ .

■ KEY INSIGHT: The y-intercept always evaluates at  $x = 0$ :  $g(0) = f(0) + 7$ .

■ FAST METHOD:  $g(0) = f(0) + 7 = -4 + 7 = 3$ .

■ 🏠 VERDICT: Hand-solve — direct substitution.

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### Question 18 [Hard]

Correct Answer: A) Left 2, reflect over x-axis, up 5

#### Step-by-Step Solution:

- Starting from  $f(x) = x^3$ :
- Replace  $x$  with  $(x + 2)$ : shift left 2  $\rightarrow (x + 2)^3$
- Negate: reflect over x-axis  $\rightarrow -(x + 2)^3$
- Add 5: shift up 5  $\rightarrow -(x + 2)^3 + 5 = g(x)$  ✓

#### Why each wrong option fails:

- B) Right 2 —  $(x + 2)$  means left, not right.
- C) Down 5 — +5 means up.
- D) Right 2 — Wrong horizontal direction.

■ ■ TRAP ALERT: Remember:  $(x + 2)$  shifts LEFT. The negative in front reflects over x-axis, not y-axis.

■ KEY INSIGHT: Read transformations from the expression: inside affects horizontal (opposite), outside affects vertical (same).

■ FAST METHOD:  $x + 2$  = left 2, negative = x-axis flip, +5 = up 5.

■ 🏠 VERDICT: Hand-solve — reading transformations from the equation.

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### Question 19 [Hard]

Correct Answer: 3

Step-by-Step Solution:

- $f(x) = |2x - 6| - 3 = |2(x - 3)| - 3$ .
- The absolute value reaches its minimum (0) when  $2(x - 3) = 0$ , i.e.,  $x = 3$ .
- Minimum of  $f$ :  $|0| - 3 = -3$ , occurring at  $x = 3$ .

■ TRAP ALERT: Students might set  $2x - 6 = -3$  instead of  $2x - 6 = 0$  to find the minimum.

■ KEY INSIGHT: The minimum of  $|expression| - c$  is  $-c$ , occurring when  $expression = 0$ .

■ FAST METHOD: Set the inside of the absolute value = 0:  $2x - 6 = 0 \rightarrow x = 3$ .

■ VERDICT: Hand-solve — set inside of absolute value to zero.

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### Question 20 [Hard]

Correct Answer: A) [-5, 7]

Step-by-Step Solution:

- $f$  has range  $[-3, 9]$ .
- $-f(2x)$  negates the range:  $[-9, 3]$  (flip the interval).
- $-f(2x) + 4$  shifts up 4:  $[-9 + 4, 3 + 4] = [-5, 7]$ .
- Note:  $f(2x)$  compresses horizontally but doesn't change the range.

Why each wrong option fails:

- B)  $[-9, 3]$  — Forgot the +4 shift.
- C)  $[-5, 3]$  — Only shifted one endpoint.
- D)  $[-13, 7]$  — Incorrectly applied the negative.

■ TRAP ALERT: When negating a range  $[a, b]$ , it becomes  $[-b, -a]$ , not  $[-a, -b]$ .

■ KEY INSIGHT: Negate:  $[a, b] \rightarrow [-b, -a]$ . Then shift up  $k$ :  $[-b+k, -a+k]$ .

■ FAST METHOD: Range of  $f$ :  $[-3, 9]$ . Negate:  $[-9, 3]$ . Add 4:  $[-5, 7]$ .

■ VERDICT: Hand-solve — systematic range transformation.

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## ■ TOPIC SUMMARY

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- 1. VERTICAL SHIFTS:**  $f(x) + k$  shifts UP  $k$ .  $f(x) - k$  shifts DOWN  $k$ .
- 2. HORIZONTAL SHIFTS:**  $f(x - h)$  shifts RIGHT  $h$ .  $f(x + h)$  shifts LEFT  $h$ . (Counterintuitive signs!)
- 3. REFLECTIONS:**  $-f(x)$  reflects over x-axis.  $f(-x)$  reflects over y-axis.
- 4. VERTICAL STRETCH/COMPRESS:**  $af(x)$  stretches vertically by factor  $|a|$ . Multiplies all y-values.
- 5. HORIZONTAL STRETCH/COMPRESS:**  $f(bx)$  compresses horizontally by factor  $|b|$ . Divides x-values.
- 6. POINT MAPPING:** Point  $(a, b)$  on  $f$  maps to  $(a + h, kb + c)$  on  $kf(x - h) + c$ .
- 7. DOMAIN vs RANGE:** Horizontal changes affect DOMAIN. Vertical changes affect RANGE.
- 8. VERTEX FORM:**  $a(x - h)^2 + k$ . Vertex at  $(h, k)$ . Min/max value is  $k$ .
- 9. ABSOLUTE VALUE:**  $|x - h| + k$  has vertex at  $(h, k)$ . Minimum of  $|expression|$  is 0.
- 10. DESMOS vs HAND:** Hand-solve for simple shifts/reflections. Desmos for complex multi-step transformations to verify.